Slick Trace investigation report — case_0003

Generated 2026-09-25T18:33:48.143100+00:00 · reproducible investigation report

What this document is not. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

Finding

No vessel in this scene can account for the observed slick. No candidate release reproduced the observation above the absolute fit gate. The system returns no name rather than the least-bad one; a source vessel may have left the scene before acquisition.

Scene

Field	Value
Incident	case_0003
Acquired (UTC)	2026-09-10T04:30:00+00:00
Sensor	synthetic SAR-like (VV proxy, single channel)
Data status	synthetic
Pixel spacing	433.59 m
Label source	generator ground truth (synthetic) — not a human or independently confirmed label

Evidence layers

Layer	Provenance
AIS	synthetic (fictional MMSIs; no real broadcast data)
Forcing	slicktrace_core.synth_generator constant+spatial field v0.2
Licence	Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply.

Traffic filtering

State	Vessels
unassessed	3
candidate	2

Unassessed is reported separately from excluded: missing evidence is not exoneration, and collapsing the two would silently convert one into the other.

Rank fragility

Leader retained in 0.7 of 80 tested perturbations. A tested perturbation changed the leader; the smallest is recorded in the JSON report.

Absence of a flip is evidence of stability across the tested set, not a proof of stability: single-axis and two-axis perturbations only.

File integrity

File	SHA-256 (first 16)
attribution_result.json	182cfa47167af72f
coverage_result.json	3d64094b061ffdd1
forcing.npz	767ec8f532e8cc46
forecast_result.json	f02d0ac96b10581a
fragility_result.json	251152903d437682
historical_mapping.json	89ae17563f55b780
manifest.json	ac936583396881e2
mask_groundtruth.png	a9dedae51c4cf6cb
naive_baseline_result.json	fedf1188f395724a
next_observation_result.json	c12e2e32d9f6468a
scene.png	4aec948e59525470

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

Uncertainty

• Rankings express physical consistency under tested assumptions. They are not proof of responsibility.

• Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.

• Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.

• Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

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